

CSDS 302 Discrete Math | Fall 2026 | Homework 4

Instructions

Section 4.5: 14, 21, 24, 38

Section 4.6: 7, 19, 20

Section 4.7: 4, 6, 17

Section 4.8: 14, 23

Section 4.10: 15, 16

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Feedback

Score: 100%

Good work.

HW4

Section 4.5

(14) Theorem
 $1000/7 = 142$ full weeks
 remainder 6 $\Rightarrow$ Monday

(21) $b \in \mathbb{Z}$ $b \bmod 12 = 5$
 what is $8b \bmod 12$
 $b = 12n + 5$ $n \in \mathbb{Z}$
 $8b = 8(12n + 5)$
 $8b = 96n + 40$
 $8b = 96n + 4$
 $8b \bmod 12 = 4$

(24) $\forall m, n \in \mathbb{Z}$
 $m \bmod 5 = 2$ $n \bmod 5 = 1$
 $mn \bmod 5 = 2$
 $m = 5k + 2$ where $k \in \mathbb{Z}$
 $n = 5q + 1$ $q \in \mathbb{Z}$
 $m \cdot n = (5k + 2)(5q + 1) = 25kq + 5k + 10q + 2$
 $m \cdot n = 5(5kq + k + 2q) + 2$
 $\underbrace{\hspace{10em}}_{\in \mathbb{Z}}$
 $\downarrow$
 $mn \bmod 5 = 2$

(38) $\forall m \in \mathbb{Z}$ $m^2 = 5k$ $\vee$ $m^2 = 5k + 1$ $\vee$ $m^2 = 5k + 4$ $k \in \mathbb{Z}$
 Assume:
 $m = 5k$ (divisible by 5)
 $m^2 = 25k = 5(5k)$ 0 remainder

$m = 5k + 1$ (like 6, 11)
 $m^2 = 25k^2 + 10k + 1 = 5(5k^2 + 2k) + 1$

$m = 5k + 2$
 $m^2 = 25k^2 + 20k + 4 = 5(5k^2 + 4k) + 4$

$m = 5k + 3$
 $m^2 = 25k^2 + 30k + 9 = 5(5k^2 + 6k + 1) + 4$

$m = 5k + 4$
 $m^2 = 25k^2 + 40k + 16 = 5(5k^2 + 8k + 3) + 1$

Section 4.6

(7) $k \in \mathbb{Z}$ $\lceil k + \frac{1}{2} \rceil$
 $\lceil k + \frac{1}{2} \rceil = k + 1$ because it rounds up
 $\lceil k + \frac{1}{2} \rceil = \lceil k + 0.5 \rceil$ & k is an int

(9) $\forall x \in \mathbb{R}$ $\lceil x - 1 \rceil = \lceil x \rceil - 1$
 $\lceil x - 1 \rceil \Rightarrow x - 1 + 1 = x$ $\lceil 2 - 1 \rceil = 2$
 $\lceil x \rceil - 1 = x$ $\lceil 2 \rceil - 1 = 2$
 TRUE

(20) $\forall x, y \in \mathbb{R}$ $\lceil xy \rceil = \lceil x \rceil \cdot \lceil y \rceil$
 let $x = 1$, $y = 2$
 $\lceil 1 \cdot 2 \rceil = 3$ $\lceil 1 \rceil \cdot \lceil 2 \rceil = 2 \cdot 3 = 6$
 Therefore false

4.7

(4) $\forall m, 7m + 4 \nmid 7$
 let's say
 $7m + 4 \mid 7$ then
 $7(m + \frac{4}{7})$, but $m + \frac{4}{7} \notin \mathbb{Z}$, false
 Therefore $7m + 4 \nmid 7$

(6) key: there is a greatest negative number
 let's say n is the greatest neg number
 $n < 0$

but let's say there's a number $m \in \mathbb{R}$
 $m = \frac{n}{2} \Rightarrow$ now m is the greatest
 $\mathbb{R}$ number, $\therefore$ FALSE

(17) a, b, c are prime
 $a^2 + b^2 \neq c^2$
 Let's say $a^2 + b^2 = c^2$
 $a^2 = c^2 - b^2 = (c-b)(c+b)$
 both can't be prime
 therefore true

4.8

(14) $n, m \in \text{irrational}$

$$n = \pi \quad m = 4 - \pi$$

$$n + m = \pi + (4 - \pi) = 4 \quad \text{which is rational}$$

FALSE

(23) $\forall a \in \mathbb{Z} \quad a(a^2 - 3) \quad \exists a \in \mathbb{Z}, a/a^2 \cdot 3$

$$a^2 = 9c + 3$$

$$a^2 = 3(c+1)$$

$$3/a^2 \Rightarrow \frac{3}{a^2} \Rightarrow \frac{3^2}{a^2} = \frac{9}{a^2}$$

$$\text{since } 3 < 9 \quad 9 \nmid a^2 - 3$$

Statement is true

4.10

(15) $a = 10993 \quad b = 832$
 $832 \neq 0$

$$R1: 10993 \bmod 832 = 117$$

$$a = 832 \quad b = 117$$

$$R2: 117 \neq 0 \quad 832 \bmod 117 = 13$$

$$a = 117 \quad b = 13$$

$$R3: 13 \neq 0 \quad 117 \bmod 13 = 0$$

$$\gcd(10993, 832) = 13$$

(16) $a = 4131 \quad b = 2431$

$$R1: 2431 \neq 0$$

$$4131 \bmod 2431 = 1700$$

$$a = 2431 \quad b = 1700$$

$$R2: 1700 \neq 0$$

$$2431 \bmod 1700 = 731$$

$$a = 1700 \quad b = 731$$

$$R3: 731 \neq 0$$

$$1700 \bmod 731 = 238$$

$$a = 731$$

$$b = 238$$

$$R4: 731 \bmod 238 = 17$$

$$a = 238 \quad b = 17$$

$$R5: 17 \neq 0$$

$$238 \bmod 17 = 0$$

$$\gcd(4131, 2431) = 17$$